**Computational MR imaging**
**Laboratory 1: MRI pulse sequences and image contrast**

Code submission is due by 23:55 on Monday. Please upload your code to StudOn in a described format. Late submissions will not be accepted.

**Learning objectives**

- Recall and improve familiarity of the MR contrast generation process.
- Recall and improve familiarity with MR pulse sequences and the k-space concept.
- Simulate image contrasts from generated by pulse sequences using a numerical brain phantom.

**1. Inspect the digital brain phantom (Aubert-Broche, Neuroimage 2006)**

1.1. Load the file *digital_brain_phantom.mat* by calling *load_data()* method. The phantom contains simulated T1, T2, and PD values. All pixels within the same region share the same value.

1.1.1. Included data

- **Label** (mat['ph']['label'][0][0])
  - 1: Cerebrospinal Fluid (CSF)
  - 2: Gray Matter (GM)
  - 3: White Matter (WM)
- **T1_map** (mat['ph']['t1'][0][0])
  - Predefined T1 values
- **T2_map** (mat['ph']['t2'][0][0])
  - Predefined T2 values
- **PD_map** (mat['ph']['sd'][0][0])
  - Predefined proton density(PD) values

1.2. Display the three tissue regions with the following methods: *get_[csf, gm, wm]_mask()*

- CSF (label=1)
- GM (label=2)
- WM (label=3)

1.3. Exploring T1, T2 and Proton Density (PD) values

1.3.1. Display T1_map, T2_map and PD_map

1.3.2. Print unique values of T1_map, T2_map, and PD_map.

1.3.3. Print corresponding T1, T2, and PD values for CSF, GM and WM, respectively by implementing methods: *get_[T1, T2, PD]().*

**2. Simulate MR image contrast from pulse sequences**

2.1. Simulate the Spin-Echo signal equation from Bernstein eq 14.57 (See Appendix) in method: *spin_echo()*.

2.2. Spin Echo Proton density weighted (PDw), T1-weighted and T2-weighted image:
- Define suitable repetition time (TR) and echo time (TE) pairs accordingly for proton density weighted, T1-weighted, and T2-weighted images, respectively.
- TR-TE pairs should be defined and stored in the constructor:
  - Proton density weighted: self.PDw_TRTE=[TR, TE]
  - T1-weighted: self.T1w_TRTE=[TR, TE]
  - T2-weighted: self.T2w_TRTE=[TR, TE]

2.3. Plot PDw, T1w, and T2w images.

## 3. K-Space trajectories (No coding required)

Consider three popular k-space acquisition trajectories. Cartesian, radial and spiral (see **Figure 1**). Sketch the gradient waveforms that generate these k-space trajectories. Hint: Remember that:

$$k_y = \int_0^t G_y d\tau$$
$$k_x = \int_0^t G_x d\tau .$$

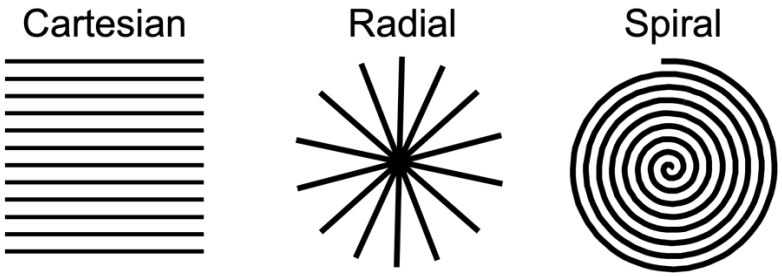



**Figure 1:** Cartesian, radial and spiral k-space trajectories

**Appendix: Equations from Bernstein book:**

Spin Echo signal equation (section 2.1.2)

$$S = M_0(1 - 2e^{-(\mathrm{TR}-\mathrm{TE}/2)/T_1} + e^{-\mathrm{TR}/T_1})e^{-\mathrm{TE}/T_2} \qquad (14.57)$$